

Uma Agrawal

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PROFESSIONAL SUMMARY

"AI-focused Software Developer with experience in building production-grade conversational AI systems using LLMs, LangChain, and LangGraph. Skilled in developing scalable AI workflows, integrating APIs, and designing intelligent systems for real-world automation including SMS-based interactions and knowledge-driven assistants."

EDUCATION

Anand International College of Engineering, Jaipur, Rajasthan Sep 2021 – Jul 2025

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence)

Marks: 92.5%

Kanha Makhan Public School, Mathura, Uttar Pradesh Jul 2019 – Apr 2020

Senior Secondary

Marks: 83%

SKILLS

Programming: Python, SQL, , HTML, CSS

AI / GenAI: RAG, LLM Applications, Prompt Engineering, LangChain, LangGraph, Conversational AI

Backend & APIs: FastAPI, REST APIs, Webhooks, Supabase Edge Functions, API Integration

AI Integrations: OpenAI, Gemini, LiveAvatar, HeyGen

Databases: PostgreSQL, SQLite, Pinecone

Tools: Git, Supabase, Environment Configuration

EXPERIENCE

Amplework Software Pvt. Ltd.

Jun 2025 – Present

Junior Associate Software Developer

- Developed production-grade AI applications using LLMs to automate customer workflows and business processes.
- Built and deployed a conversational SMS bot integrating FastAPI, LangGraph, and external backend services.
- Designed multi-agent workflows for intent classification, lead processing, and response generation using LangChain.
- Integrated APIs with .NET backend systems and CRM platforms for seamless data exchange.
- Implemented session management and conversation tracking using SQLite for multi-user interactions.
- Optimized RAG pipelines and prompt strategies to improve response relevance and reduce hallucinations.
- Managed environment configurations (Supabase, API keys, Vite) for secure deployments.

PROJECTS

Kinetic AI SMS Bot (Conversational AI System)

- Built a production-grade SMS-based AI system using FastAPI and LangGraph for automated customer interaction and appointment management.
- Designed multi-agent workflows for intent classification, lead handling, and response generation.
- Integrated Dialpad webhooks, .NET services, and CRM systems for end-to-end communication.
- Implemented session management using SQLite to maintain conversation history.
- Developed context-aware responses with memory and tool-based execution.

AI Knowledge Assistant (Advanced RAG System)

- Built a production-ready RAG system supporting multi-source data ingestion (PDFs, URLs, documents).
- Implemented semantic search using Pinecone with hybrid retrieval strategies.
- Designed conversational workflows using LangChain and LangGraph with memory handling.

- Integrated backend APIs and PostgreSQL for scalable data management.

AI Pipeline for Image Segmentation & Object Analysis

- Developed a deep learning pipeline using CNNs for image segmentation and object detection.
- Applied preprocessing techniques and evaluated performance using IoU and accuracy metrics.

CERTIFICATES AND TRAININGS

- AI Tools Workshop – Model Development & Deployment
- Data Visualization (Tata Virtual Internship): Completed insightful visual reports using tools to effectively communicate data findings.
- Machine Learning (Forsk Coding School): Applied ML algorithms to real-world datasets.
- Python (Internshala): Completed Python basic to advanced training focusing on programming skills.